6(a) $^{1}_{0}n + ^{235}_{92}U \rightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3\,^{1}_{0}n$

3 neutrons are produced.

(b) From the graph U has lower binding energy per nucleon than the products Ba and Kr. Multiplying by the number of nucleons, U has lower binding energy than both (Ba + Kr) together.
This means that U required less energy to break up the nucleus into its constituent nucleons, and more energy is being released when both Ba + Kr are formed from the constituent nucleons. Hence overall, there is a net release of energy.

(c) Energy required to break U = 7.63 x 235 = 1793 MeV
Energy released when Ba + Kr formed = 8.40 x 141 + 8.70 x 92 = 1985 MeV
Net energy released = 1985 – 1793 = 192 MeV.

(d) Neutron is a free particle and has no binding energy. Thus it is not included in the calculation.

7 (a)